Driver App Self-Learning Algorithm

Purpose

The app should feel like a useful, fast-scrolling social feed, but its recommendation system should optimize for **learning and safe decision-making**, not merely time spent in the app.

The algorithm uses four simple component formulas:

1. **Accuracy score** — how well the person answers.
2. **Work score** — how much meaningful work the person actually does.
3. **Attention score** — whether the person watches and engages with the lesson.
4. **Time-quality score** — whether the time spent suggests thoughtful work rather than guessing or getting stuck.

These four scores combine into a **mastery score**. The mastery score determines whether the learner can move forward, needs correction, or should see the topic again later.

All component scores are normalized to a range of **0.00 to 1.00**.

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1. Accuracy score: `A`

Accuracy must be the most important signal. A learner should not be able to earn mastery just by watching videos or tapping through activities.

Variables

- `C1` = first-answer correctness: `1` if correct, `0` if incorrect.
- `C2` = correction-answer correctness: `1` if correct after feedback, `0` if incorrect.
- `C3` = later-review correctness: average correctness on this topic during later review, or `0.5` when there is not enough history.

Formula

Plain Text

```
A = 0.60*C1 + 0.25*C2 + 0.15*C3
```

Why these weights?

- The first answer gets **60%** because it best shows what the learner knew before help.

- The correction answer gets **25%** because learning from feedback matters.
- Later review gets **15%** because remembering after time has passed is important.

Examples

- Correct immediately, no history: `A = 0.60*1 + 0.25*0 + 0.15*0.5 = 0.675`.
- Wrong first, correct after correction, no history: `A = 0.60*0 + 0.25*1 + 0.15*0.5 = 0.325`.
- Correct first and correct later: `A = 0.60*1 + 0.25*0 + 0.15*1 = 0.75`.

A single correct answer should not automatically mean permanent mastery. The later-review term is what makes the algorithm self-learning over time.

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2. Work score: `W`

The work score measures whether the learner actually attempted the learning task. It rewards meaningful effort, but it does **not** allow effort to replace correctness.

Variables

- `Q` = question completion: `1` if the learner submits an answer, `0` if they abandon it.
- `R` = retry completion: `1` if they complete the correction or similar question when one is assigned, otherwise `0`.
- `I` = interactive activity completion: `1` if they complete the assigned activity, otherwise `0`.
- `H` = help use: `1` if they open the explanation/help, otherwise `0`.

Formula

Plain Text

```
W = 0.40*Q + 0.30*R + 0.20*I + 0.10*H
```

If a component is not assigned, remove it and divide by the weights that were actually applicable. This prevents a lesson without an interactive activity from unfairly lowering the score.

Example

The learner answers the original question, completes the correction question, skips the interactive activity, and opens help:

Plain Text

```
W = 0.40*1 + 0.30*1 + 0.20*0 + 0.10*1
W = 0.80
```

The learner did meaningful work, but the work score does not hide whether the answers were correct.

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3. Attention score: `N`

Attention measures whether the learner meaningfully consumed the lesson rather than instantly scrolling past it.

Variables

- `V` = video completion ratio, capped at `1.00`.
  - Example: watched 80% of video → `V = 0.80`.
- `K` = key-part completion: `1` if the required important segment was watched, otherwise `0`.
- `E` = active engagement: fraction of required purposeful actions completed, such as tapping a hazard, choosing right-of-way, or matching a sign.

Formula

Plain Text

```
N = 0.50*V + 0.30*K + 0.20*E
```

Why include the key part?

A learner may watch most of a video while missing the specific rule the quiz tests. The key-part signal lets the app say, “Watch the important part,” without requiring the person to replay the entire video.

Example

- Watched 100%: `V = 1.00`.
- Watched the required rule segment: `K = 1`.
- Completed half of the optional interactive challenge: `E = 0.50`.

Plain Text

```
N = 0.50*1.00 + 0.30*1 + 0.20*0.50 = 0.90
```

The app should track video progress and meaningful interaction, not pretend that raw screen-open time equals attention.

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4. Time-quality score: `T`

Time should be used carefully. The app should not reward someone merely for spending a long time staring at a screen, and it should not punish a fast learner for answering quickly.

Time is therefore a **small diagnostic signal**, not a major grade.

Variables

- `t` = active seconds spent on the question, excluding background time.
- `b` = expected healthy time for that question type. This can start as a simple fixed value, such as `25` seconds for a standard multiple-choice question.
- `r = t / b` = time ratio.

Formula

Plain Text

```
if 0.50 <= r <= 2.00: T = 1.00
if 0.25 <= r < 0.50: T = 0.70
if r < 0.25: T = 0.45
if 2.00 < r <= 4.00: T = 0.75
if r > 4.00: T = 0.55
```

Interpretation:

- Very fast answers may be guesses or accidental taps.
- Normal-time answers receive full time quality.
- Long answers may indicate careful thought or confusion, so they receive a moderate score rather than an automatic failure.
- The app should not use time alone to lock or shame a learner.

Example

Expected time is `25` seconds and the learner spends `18` active seconds:

Plain Text

```
r = 18 / 25 = 0.72
T = 1.00
```

If the learner spends only `4` seconds:

Plain Text

```
r = 4 / 25 = 0.16
T = 0.45
```

That low time score tells the algorithm to check the topic again, but it does not say the learner is unintelligent or failed.

Mastery score: `M`

The four components combine as follows:

Plain Text

```
M = 0.55*A + 0.15*W + 0.20*N + 0.10*T
```

Why these weights?

- **55% Accuracy:** the learner must demonstrate correct knowledge.
- **15% Work:** completing correction and practice matters.
- **20% Attention:** the learner needs exposure to the rule and its explanation.
- **10% Time quality:** time can identify guessing or confusion, but should not dominate.

The weights sum to 1.00, so `M` remains between `0.00` and `1.00`.

Mastery bands

Mastery score	Meaning	App behavior
`0.80–1.00`	Strong evidence of mastery	Allow learner to move on; schedule normal spaced review.
`0.60–0.79`	Developing understanding	Allow movement after a short reinforcement or similar question.

`0.40–0.59`	Needs support	Show a correction, then require a similar question or short activity.
`0.00–0.39`	High need for instruction	Show the important video segment, correction, and interactive practice before moving on.

These bands are not “grades.” They are routing instructions for the app.

Worked examples

Example A: Learner already knows the topic

The learner watches the lesson, answers correctly immediately, and completes no correction because none is needed.

Plain Text

```
A = 0.675       first answer correct, no later history yet
W = 0.40        submitted the question
N = 1.00        watched the full video
T = 1.00        answered in a healthy time range

M = 0.55(0.675) + 0.15(0.40) + 0.20(1.00) + 0.10(1.00)
M = 0.69125 ≈ 0.69
```

This is in the developing range because the learner has not yet proved retention. The app can give a harder confirmation question. If that is correct, increase the topic's later-review score and allow the learner to move on quickly.

Example B: Learner watches but misses the rule

The learner watches the full video, answers incorrectly, opens help, completes the correction question correctly, and spends a normal amount of time.

Plain Text

```
A = 0.325       wrong first answer, correct correction, no later history
W = 0.80        original + correction + help completed
N = 1.00        full video watched
T = 1.00        healthy answer time
```

```
M = 0.55(0.325) + 0.15(0.80) + 0.20(1.00) + 0.10(1.00)
M = 0.61875 ≈ 0.62
```

The learner can move forward after the correction because the behavior shows learning, even though first-pass knowledge was weak.

Example C: Learner skips and guesses

The learner watches 20% of the video, answers in four seconds, gets it wrong, and does not complete the correction.

Plain Text

```
A = 0.075       wrong first answer, no correction, no history
W = 0.40        submitted only the original question
N = 0.40        low video completion and no key segment
T = 0.45        unusually fast answer

M = 0.55(0.075) + 0.15(0.40) + 0.20(0.40) + 0.10(0.45)
M = 0.20225 ≈ 0.20
```

The app should not say “You failed.” It should say:

> “Almost. Here is the rule that changes the answer.”

Then it should show the important segment or an interactive correction and ask a similar question.

The feed/recommendation rule

The mastery score answers **“How well does this learner know the topic?”** It should not be the only factor deciding what appears next.

For each topic, maintain:

- `M` = current mastery score.
- `D` = days since the topic was last practiced.
- `X` = number of times the learner skipped or requested review later.
- `S` = safety importance, set by curriculum authors from `0.50` to `1.00`.

Review priority

Plain Text

```
P = 0.50*(1 - M) + 0.20*min(D/7, 1) + 0.20*min(X/3, 1) + 0.10*S
```

Higher `P` means the topic should appear sooner in the daily review or feed.

This creates the social-feed-like personalization effect without copying the harmful objective of maximizing endless scrolling:

- Weak topics rise because `1-M` is high.
- Forgotten topics rise because days since practice increases.
- Skipped topics return because `X` increases.
- Safety-critical topics get a modest priority boost through `S`.

Example

Topic 1:

Plain Text

```
M = 0.35, D = 1, X = 1, S = 1.00
P = 0.50(0.65) + 0.20(1/7) + 0.20(1/3) + 0.10(1.00)
P ≈ 0.54
```

Topic 2:

Plain Text

```
M = 0.85, D = 6, X = 0, S = 0.70
P = 0.50(0.15) + 0.20(6/7) + 0 + 0.10(0.70)
P ≈ 0.27
```

Topic 1 should be shown before Topic 2, even if Topic 2 was recently popular. The system is optimizing useful review, not popularity.

Routing rules for the user experience

The formula and the interface should work together:

1. **Video skipped:** show the quiz with a clear note that the rule may matter on the permit test or in real driving. Offer `Answer now`, `Watch important part`, or `Review later`.
2. **Correct answer:** allow progress, unless the learner chose `I already know this`; then give a harder confirmation question.
3. **Score below 0.60:** show correction content and a similar question.

4. **Second incorrect answer:** show a short interactive activity or require the important video segment.
5. **Review later:** add the topic to the daily queue; never silently discard it.
6. **Daily review:** select topics using `P`, with a mix of today, yesterday, two days ago, skipped topics, and previously incorrect topics.
7. **Mastery decay:** do not permanently erase mastery. Let the priority formula bring old topics back as they become due for review.

What makes it self-learning?

The first version does not need machine learning. It is self-learning because it updates the learner model after every interaction:

```
Plain Text
new_mastery = 0.70*old_mastery + 0.30*current_session_mastery
```

This smoothing prevents one lucky guess or one bad day from changing the learner's profile too dramatically.

For each topic, save:

- `old_mastery`
- `current_session_mastery`
- `last_practiced_at`
- `attempt_count`
- `correct_count`
- `video_completion`
- `average_active_time`
- `skip_count`
- `help_count`
- `retry_count`
- `review_priority`

Later, once enough data exists, the expected time `b`, thresholds, and weights can be tuned from real outcomes. The formulas should remain visible and explainable to the product team.

Important product guardrails

- Never call a learner a failure.
- Never make time spent the main measure of learning.
- Never allow video completion to substitute for a correct answer.
- Do not create an infinite-scroll loop that rewards compulsive use; cap daily review length and let the learner stop.
- For safety-critical driving rules, a curriculum author should be able to mark a topic as high `S` and require a minimum knowledge check before completion.
- Store only the behavior needed to improve learning, with clear privacy controls and an opt-out for analytics beyond core progress tracking.